Worksheet 2.2 — Remainder & Factor Theorems

Grade 12 Advanced Functions (MHF4U) · Unit 2: Polynomial Equations and Inequalities

The Big Ideas

The remainder of $P(x) \div (x-a)$ is $P(a)$; and $x-a$ is a factor exactly when $P(a)=0$.

- Remainder Theorem: remainder $= P(a)$.
- Factor Theorem: $x-a$ is a factor $\iff P(a)=0$.
- Rational Root Theorem: $\frac{p}{q}$ with $p \mid$ constant, $q \mid$ leading coeff.

Core Concepts

The remainder theorem

The remainder of $P(x) \div (x-a)$ is exactly $P(a)$, so one substitution replaces a full division.

The factor theorem

When $P(a)=0$ the remainder is zero and $x-a$ is a **factor** the key tool for factoring higher-degree polynomials.

Finding zeros

Test factors of the constant term as candidate zeros, then divide out each factor you confirm.

Worked Examples

Example 1 — Find a remainder

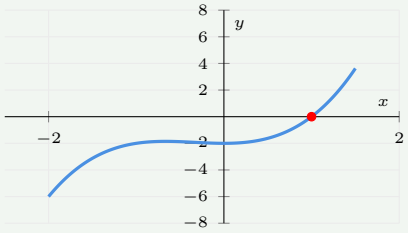
Remainder of $P(x)=x^3-4x+3$ divided by $x-3$.

Solution. $P(3)=27-12+3=18$.

Example 2 — Test a factor

Is $x-1$ a factor of x^3+x^2-2?

Solution. $P(1)=1+1-2=0$: yes. The graph crosses at $x=1$:



Example 3 — Sum of cubes

Is $x+2$ a factor of x^3+8?

Solution. $P(-2)=-8+8=0$: yes.

Example 4 — Solve for k

Find k so $x-2$ is a factor of x^3+kx-6.

Solution. $P(2)=8+2k-6=0 \Rightarrow k=-1$.

Example 5 — Find a zero

A rational zero of $P(x)=x^3-3x^2-x+3$?

Solution. Candidates $\pm1, \pm3$. $P(1)=1-3-1+3=0$, so $x=1$.

Example 6 — Remainder

Remainder of x^3+5 divided by $x-2$.

Solution. $P(2)=13$.

Example 7 — Difference of cubes

Is $x-5$ a factor of x^3-125?

Solution. $P(5)=125-125=0$: yes.

Example 8 — Solve for k

Find k so $x+2$ is a factor of $x^3+kx+10$.

Solution. $P(-2)=-8-2k+10=0 \Rightarrow k=1$.

Example 9 — Find a zero

A rational zero of $x^3-2x^2-9x+18$?

Solution. $P(2)=8-8-18+18=0$, so $x=2$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Remainder of $x^3 + 4$ divided by $x + 1$?

Question 2

Is $x - 4$ a factor of $x^3 - 64$?

Question 3

Is $x + 2$ a factor of $x^3 + 2x^2 - 4x - 8$?

Question 4

Find k so $x - 3$ is a factor of $x^3 + kx^2 - 9$.

Question 5

A rational zero of $x^3 + x^2 - 4x - 4$?

Question 6

Remainder of $x^3 - 3x + 2$ divided by $x - 1$?

Question 7

Is $x - 3$ a factor of $x^3 - 27$?

Question 8

Find k so $x - 2$ is a factor of $x^3 - kx + 2$.

Question 9

Remainder of $2x^3 - x + 5$ divided by $x + 1$?

Question 10

Is $x+3$ a factor of x^3+27?

Question 11

List the rational-root candidates for x^3-2x^2-5x+6.

Question 12

Remainder of x^4-1 divided by $x-1$?

Question 13 — Challenge

Show that $x-1$ is a factor of x^3-2x^2-x+2, then state another factor.

Question Recap & Answer Key

Question Recap

1. Remainder of $x^3 + 4$ divided by $x + 1$?
2. Is $x - 4$ a factor of $x^3 - 64$?
3. Is $x + 2$ a factor of $x^3 + 2x^2 - 4x - 8$?
4. Find k so $x - 3$ is a factor of $x^3 + kx^2 - 9$.
5. A rational zero of $x^3 + x^2 - 4x - 4$?
6. Remainder of $x^3 - 3x + 2$ divided by $x - 1$?
7. Is $x - 3$ a factor of $x^3 - 27$?
8. Find k so $x - 2$ is a factor of $x^3 - kx + 2$.
9. Remainder of $2x^3 - x + 5$ divided by $x + 1$?
10. Is $x + 3$ a factor of $x^3 + 27$?
11. List the rational-root candidates for $x^3 - 2x^2 - 5x + 6$.
12. Remainder of $x^4 - 1$ divided by $x - 1$?
13. Show that $x - 1$ is a factor of $x^3 - 2x^2 - x + 2$, then state another factor.

✓ Answer Key

1. 3
2. yes
3. yes
4. $k = -2$
5. $x = -1$
6. 0
7. yes
8. $k = 5$
9. 4
10. yes
11. $\pm 1, \pm 2, \pm 3, \pm 6$
12. 0
13. $P(1) = 0$; e.g. $(x - 2)$ or $(x + 1)$